Chapter 5: The Multiple Regression Model

Prices, advertising, and the evidence behind a decision

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Session 1: Holding other variables constant

Andy's decision

You work in the **economics department at Big Andy's Burger Barn**.

Management wants to increase monthly sales revenue. Your team has data on prices, advertising expenditure, and sales from 75 cities.

You have been asked:

Should we reduce our prices or spend more on advertising?

- Predict the sign of each relationship with sales revenue.
- What evidence would you need before giving advice?

Prices and sales: a first look

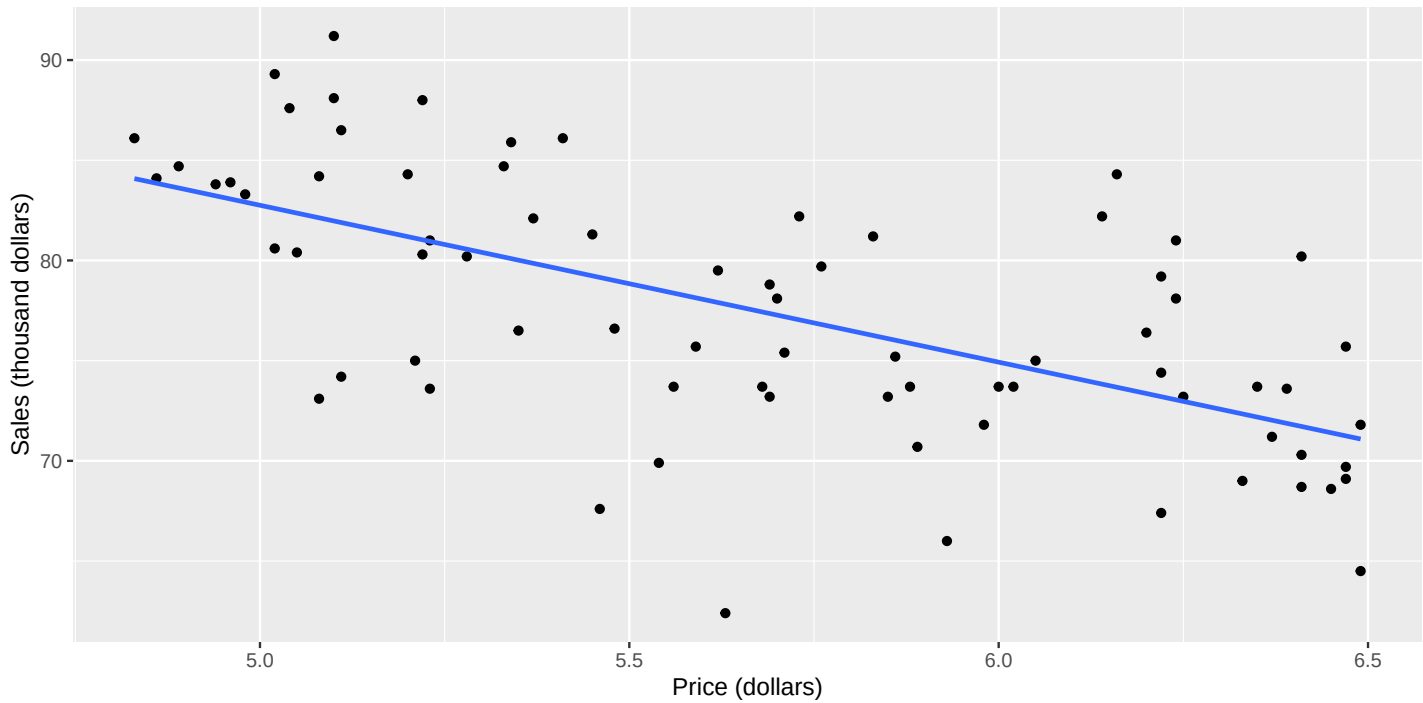


Figure 1: Sales and price, with a simple OLS regression line

What does the line suggest about price and sales revenue?

R script · Part 1: Price plot

Advertising and sales: a first look

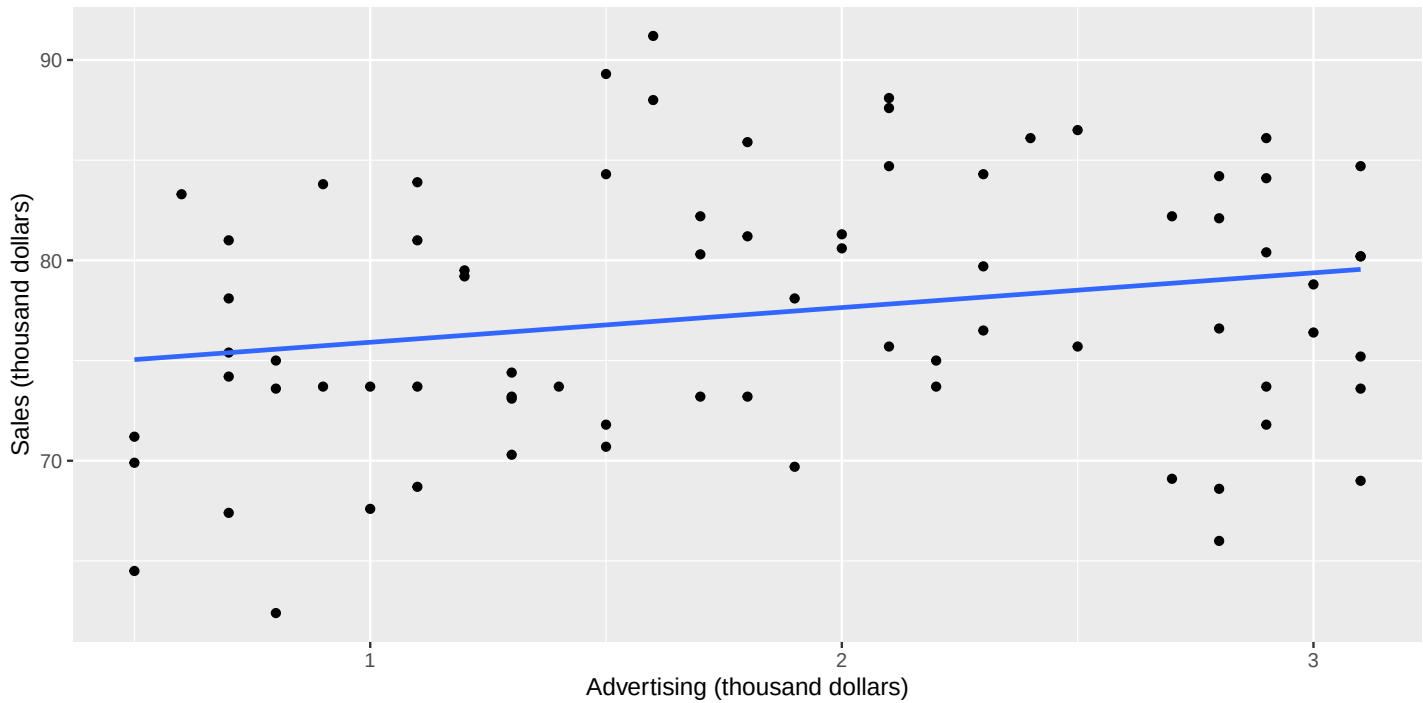


Figure 2: Sales and advertising, with a simple OLS regression line

Does this line tell us what happens when price is held constant?

R script · Part 1: Advertising plot

From one explanatory variable to two

Each graph describes sales using **one explanatory variable**.

$$SALES_i = \beta_1 + \beta_2 PRICE_i + e_i$$

If advertising varies across cities, where does its influence go?

Our next model includes price **and** advertising together.

R script · Part 1: Compare the two simple regressions

What we will be able to do

- Interpret a coefficient with other explanatory variables held constant.
- Calculate revenue changes and assess their uncertainty.
- Interpret marginal effects in models with squares and interactions.

Prior knowledge: fitted values, residuals, OLS, confidence intervals, and hypothesis tests from Chapters 2–4.

The data and their units

75 cities with Big Andy's Burger Barn restaurants.

Table 1: Andy data, textbook Example 5.1

Variable	Meaning	Unit
`sales`	Monthly sales revenue	Thousand dollars
`price`	Price index	Dollars
`advert`	Monthly advertising expenditure	Thousand dollars

Sales revenue is not profit. Production costs are not observed here.

R script · Part 1: Data and units

The multiple regression model

$$SALES_i = \beta_1 + \beta_2 PRICE_i + \beta_3 ADVERT_i + e_i \tag{1}$$

If $E(e_i \mid \mathbf{X}) = 0$:

$$E(SALES_i \mid \mathbf{X}) = \beta_1 + \beta_2 PRICE_i + \beta_3 ADVERT_i$$

or

$$E(SALES_i \mid PRICE_i, ADVERT_i) = \beta_1 + \beta_2 PRICE_i + \beta_3 ADVERT_i$$

e_i captures influences on sales that the model does not include.

In the book's general notation: $y_i = SALES_i$, $x_{i2} = PRICE_i$, and $x_{i3} = ADVERT_i$.

From a line to a plane

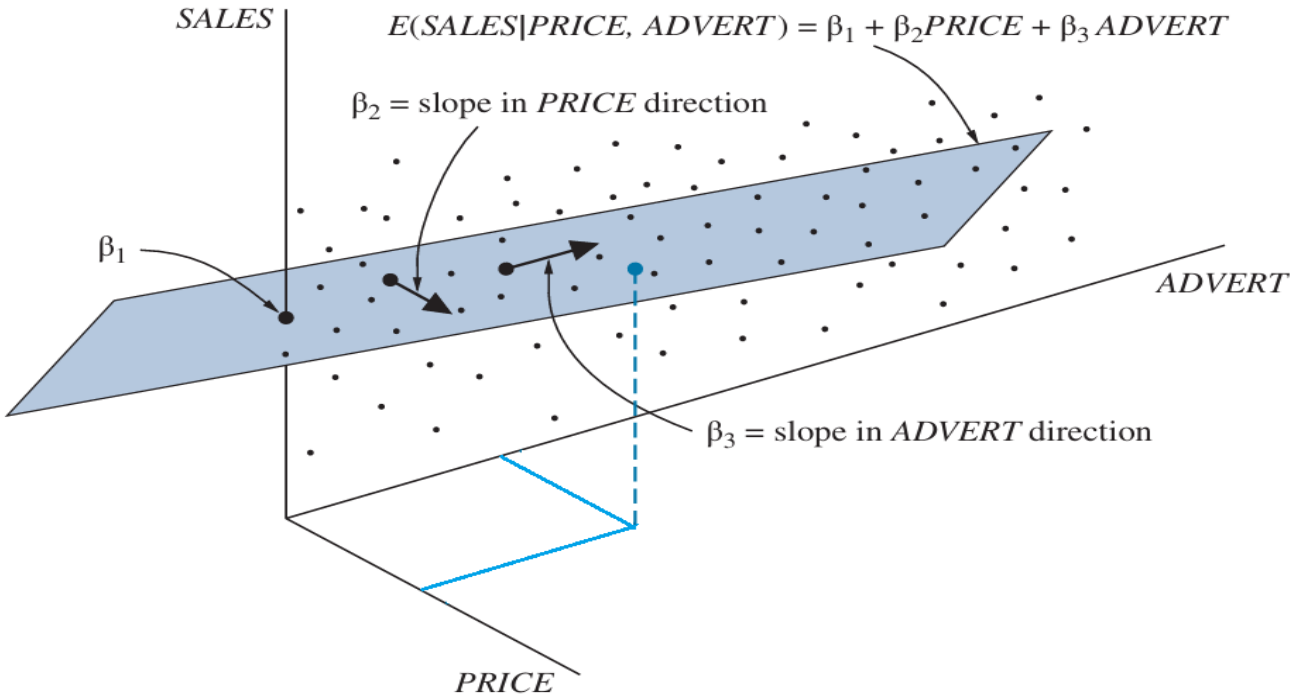


FIGURE 5.1 **The multiple regression plane.**

Figure 3: The multiple regression plane, reproduced from textbook Figure 5.1.

Holding advertising constant

Compare two settings with the **same advertising expenditure**:

$PRICE_1 = PRICE_0 + \Delta PRICE$; advertising remains $ADVERT_0$.

$$y_0 = \beta_1 + \beta_2 PRICE_0 + \beta_3 ADVERT_0$$
$$y_1 = \beta_1 + \beta_2 PRICE_1 + \beta_3 ADVERT_0$$

Subtract the first expression from the second:

$$y_1 - y_0 = _____$$

R script · Part 2: Interpret the price coefficient

What allows a causal interpretation?

$$E(e_i \mid \mathbf{X}) = 0$$

After conditioning on the included variables, the remaining influences must have zero conditional mean.

Suppose a competitor responds whenever Andy changes his price.

Can the price coefficient isolate Andy's price effect if the competitor's price is omitted?

OLS fits the plane

Choose coefficients that minimise the squared vertical distances:

$$
\begin{aligned}
S(\beta_1, \beta_2, \beta_3) &= \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \\
&= \sum_{i=1}^{N} (y_i - \beta_1 - \beta_2 x_{i2} - \beta_3 x_{i3})^2
\end{aligned}
\tag{2}
$$

```
model <- lm(sales ~ price + advert, data = andy)
```

Intuition: choose one plane that keeps the overall squared errors small.

R script · Part 2: Estimate the model

The estimated coefficients

Table 2: OLS estimates and conventional standard errors

Term	Estimate	SE
Intercept	118.914	6.352
Price	-7.908	1.096
Advertising	1.863	0.683

With advertising fixed, a 10-cent price reduction corresponds to an estimated revenue increase of **791 dollars**.

With price fixed, an additional 1,000 dollars of advertising corresponds to **1863 dollars** more revenue.

R script · Part 2: Coefficients and revenue changes

A prediction by hand

Price: 5.50 dollars. Advertising: 1,200 dollars.

$$\widehat{SALES} = 118.914 - 7.9079(5.50) + 1.8626(1.20)$$

```
f_model <- makeFun(model)
f_model(price = 5.50, advert = 1.20) * 1000
```

Expected monthly revenue: **77656 dollars**.

R script · Part 3: Predict sales with makeFun()

One residual, then all residuals

$$\hat{e}_i = SALES_i - \widehat{SALES}_i \tag{3}$$

For the first city:

$$\hat{e}_1 = 73.2 - 76.339 = -3.139$$

The city is 3.139 thousand dollars **below** the fitted plane.

Sum of squared errors: mathematics and R

$$SSE = \sum_{i=1}^{N} \hat{e}_i^2 = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{4}$$

```
e_hat <- residuals(model)
SSE <- sum(e_hat^2)
```

$SSE = 1718.943$ in squared thousand-dollar units.

Intuition: squaring prevents cancellation and gives greater weight to large residuals.

R script · Part 4: Sum of squared errors

Error variance and degrees of freedom

$$\hat{\sigma}^2 = \frac{SSE}{N-K} = \frac{1718.943}{75-3} = 23.874 \tag{5}$$

```
K <- length(coef(model))
sigma2_hat <- SSE / df.residual(model)
```

K counts **all coefficients, including the intercept**.

R script · Part 4: Error variance and degrees of freedom

Explained and unexplained variation

SST: total sum of squares. It measures variation in sales around the sample mean, before using price or advertising to predict sales.

$$SST = \sum_{i=1}^{N} (SALES_i - \overline{SALES})^2$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}, \qquad SST = SSR + SSE \tag{6}$$

SSR: explained sum of squares. **SSE:** unexplained sum of squares.

R script · Part 4: SST and R-squared

What does 44.8% explained mean?

```
SST <- sum((andy$sales - mean(andy$sales))^2)
R_squared <- 1 - SSE / SST
```

$R^2 = 0.448$: the model explains about 44.8% of the sample variation in sales around its mean.

Before fitting a model, sales already differed across cities. Using only the mean would leave all these deviations unexplained.

Price and advertising account for about **44.8%** of that variation; about **55.2%** remains unexplained.

We can explain some variation, but not all of it.

What does controlling for price actually do?

We want to compare advertising and sales **after accounting for price differences**.

- **Advertising:** is this city advertising more or less than its price predicts?
- **Sales:** are its sales higher or lower than its price predicts?

We then ask whether these two departures are related.

R script · Part 5: Holding price constant

Controlling for price: the three regressions

Remove the linear relationship with price from **both** variables:

```
sales_net_price <- residuals(lm(sales ~ price, data = andy))
advert_net_price <- residuals(lm(advert ~ price, data = andy))
fwl_model <- lm(sales_net_price ~ advert_net_price - 1)
```

The last regression uses variation remaining after price and the intercept have been removed.

R script · Part 5: The three FWL regressions

The partial relationship

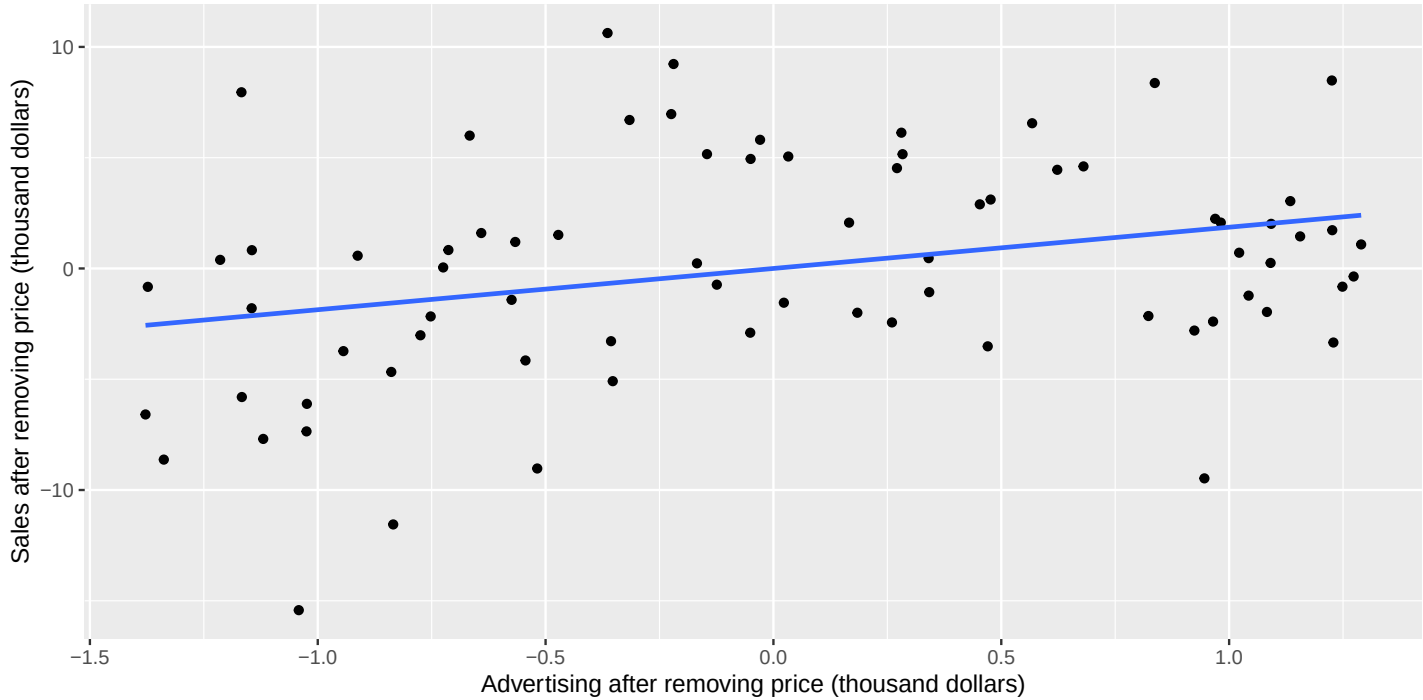


Figure 4: Sales and advertising after removing their linear relationships with price

Slope: **1.8626**, identical to the advertising coefficient in the full model.

R script · Part 5: The partial relationship plot

The assumptions behind the results

We now examine **MR1–MR6 one at a time**.

Each condition answers a different question about the model, the unobserved influences on sales, or the information in our data.

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + e_i$$

$y_i = SALES_i$, $x_{i2} = PRICE_i$, $x_{i3} = ADVERT_i$.

The model is **linear in the coefficients**.

It can include a squared variable such as $ADVERT_i^2$ and still be linear in the coefficients.

MR2: Strict exogeneity

$$E(e_i \mid \mathbf{X}) = 0$$

Given the included regressors, the remaining influences on sales have **zero conditional mean**.

Andy example: omitted competitor prices may affect sales and be related to Andy's own price.

MR3: Conditional homoskedasticity

$$\text{var}(e_i \mid \mathbf{X}) = \sigma^2$$

The spread of sales around the regression plane is constant across price and advertising settings.

What if prediction errors are much more variable at high advertising levels?

MR4: Conditionally uncorrelated errors

$$\text{cov}(e_i, e_j \mid \mathbf{X}) = 0, \qquad i \neq j$$

An unobserved sales shock in one city does not systematically move together with the error in another city, conditional on the regressors.

Could a common local demand shock affect several nearby restaurants?

MR5: No exact linear relationship among regressors

No explanatory variable can be written as an **exact linear combination** of the others, including the constant.

Suppose every city satisfies:

$$ADVERT_i = 2\,PRICE_i$$

Can we separately identify the price and advertising coefficients?

MR6: Conditionally normal errors

$$e_i \mid \mathbf{X} \sim N(0, \sigma^2)$$

At given regressor settings, errors around the regression plane are normally distributed.

With MR1–MR5, this gives **exact small-sample** t inference.

Normality is optional for the Gauss–Markov result.

Before the break

An adviser says:

> *"We included advertising, so the price coefficient must be causal."*

What is missing from this argument?

Then explain why a negative price coefficient does not justify reducing the price all the way to zero.

Session 2: From estimates to a decision

Is a positive estimate enough?

Estimated advertising coefficient: **1.863**.

An additional 1,000 dollars of advertising corresponds to about 1863 dollars more revenue.

Would you advise Andy to spend more?

What could uncertainty change about your recommendation?

R script · Review of Part 2: Advertising coefficient

Sample correlation

$$r_{23} = \frac{\sum_i (x_{i2} - \bar{x}_2)(x_{i3} - \bar{x}_3)}{\sqrt{\sum_i (x_{i2} - \bar{x}_2)^2 \sum_i (x_{i3} - \bar{x}_3)^2}}$$

How much does **price and advertising vary together** relative to their separate variation?

r_{23} ranges from -1 to 1; r_{23}^2 ranges from 0 to 1.

It is also the R^2 from a **simple regression of price on advertising** with an intercept.

It measures how much of the price variation is linearly shared with advertising.

R script · Part 6: Correlation between price and advertising

What determines precision?

For a model with price and advertising:

$$\text{var}(b_2 \mid \mathbf{X}) = \frac{\sigma^2}{(1 - r_{23}^2)\sum_{i=1}^{N}(PRICE_i - \overline{PRICE})^2} \tag{7}$$

$x_2 = PRICE$, $x_3 = ADVERT$; r_{23} is their sample correlation.

- More noise makes the estimate less precise.
- More independent price variation makes it more precise.
- Strong price-advertising correlation makes separate effects harder to identify.

Independent variation and precision

In the variance formula, the denominator is

$$\underbrace{(1-r_{23}^2)}_{\text{share remaining after advertising}} \underbrace{\sum_i (PRICE_i - \overline{PRICE})^2}_{\text{total price variation}}.$$

- If r_{23}^2 is near 0, most price variation remains available.
- If r_{23}^2 is near 1, little remains to identify a separate price effect.

For Andy: $r_{23} = 0.026$ and $r_{23}^2 = 0.0007$.

R script · Part 6: Correlation between the regressors

The coefficient covariance matrix

For b_1 (intercept), b_2 (price), and b_3 (advertising):

$$\begin{pmatrix} \widehat{\text{var}}(b_1) & \widehat{\text{cov}}(b_1, b_2) & \widehat{\text{cov}}(b_1, b_3) \\ \widehat{\text{cov}}(b_2, b_1) & \widehat{\text{var}}(b_2) & \widehat{\text{cov}}(b_2, b_3) \\ \widehat{\text{cov}}(b_3, b_1) & \widehat{\text{cov}}(b_3, b_2) & \widehat{\text{var}}(b_3) \end{pmatrix}$$

Diagonal: each coefficient estimate's variance.

Off diagonal: covariances between coefficient estimates.

Andy's estimated covariance matrix

Rows and columns: **intercept, price, advertising**.

$$\begin{pmatrix} 40.3433 & -6.7951 & -0.7484 \\ -6.7951 & 1.2012 & -0.0197 \\ -0.7484 & -0.0197 & 0.4668 \end{pmatrix}$$

```
vcov(model)
```

For advertising: $se(b_3) = \sqrt{0.4668} = 0.6832$.

R script · Part 6: vcov(model)

Standard errors describe sampling uncertainty

$$se(b_k) = \sqrt{\widehat{\text{var}}(b_k)}$$

```
V <- vcov(model)
sqrt(diag(V))
```

Advertising SE: **0.6832**.

Another sample would generally produce another coefficient estimate.

R script · Part 6: The advertising standard error

A confidence interval for advertising

$$b_3 \pm t_{0.975, N-K} \, se(b_3) \tag{8}$$

$$1.8626 \pm 1.9935 \times 0.6832$$

```
confint(model, "advert", level = 0.95)
```

95% interval: **[0.501, 3.225]**.

R script · Part 6: Confidence intervals

Two different economic claims

Table 3: Different claims require different tests

Question	Null hypothesis	Alternative
Is advertising related to sales?	$\beta_3 = 0$	$\beta_3 \neq 0$
Does extra revenue cover advertising expenditure?	$\beta_3 \leq 1$	$\beta_3 > 1$

Predict: could the data support the first claim but leave the second unresolved?

R script · Part 7: Two questions, two tests

The tests give different answers

$$t_0 = \frac{b_3}{se(b_3)}, \qquad t_1 = \frac{b_3 - 1}{se(b_3)}$$

Table 4: Advertising tests using 72 residual degrees of freedom

Claim	t	p
Advertising is related to sales	2.726	0.008
Revenue covers advertising cost	1.263	0.105

At 5%, reject the zero-effect null; do not reject the cost-coverage null.

A significant coefficient does not establish profitable advertising.

R script · Part 7: Compare the two p-values

The test calculations in R

```r
b3 <- coef(model)["advert"]
se_b3 <- sqrt(vcov(model)["advert", "advert"])
df <- df.residual(model)

t0 <- b3 / se_b3
2 * pt(-abs(t0), df = df)

t1 <- (b3 - 1) / se_b3
pt(t1, df = df, lower.tail = FALSE)
```

The first probability uses **both tails**; the second uses the **upper tail**.

R script · Part 7: Calculate the test statistics

Changing price and advertising together

Andy cuts price by 40 cents and increases advertising by 800 dollars.

$$\begin{aligned} \lambda &= E(SALES_1) - E(SALES_0) \\ &= -0.40\beta_2 + 0.80\beta_3 \end{aligned} \tag{9}$$

Before calculating: why does the intercept disappear?

$$\widehat{\lambda} = -0.40b_2 + 0.80b_3 = 4.6532$$

R script · Part 8: Change price and advertising

The covariance term matters

$$\widehat{\text{var}}(\hat{\lambda}) = (-0.40)^2\widehat{\text{var}}(b_2) + (0.80)^2\widehat{\text{var}}(b_3) + 2(-0.40)(0.80)\widehat{\text{cov}}(b_2, b_3) \tag{10}$$

```
policy_variance <- (-0.40)^2 * var_price + 0.80^2 * var_advert +
  2 * (-0.40) * 0.80 * cov_price_advert
sqrt(policy_variance)
```

$se(\hat{\lambda}) = 0.7096.$

R script · Part 8: Variance of the combined change

An interval for the combined change

$$\widehat{\lambda} \pm t_{0.95,72}\, se(\widehat{\lambda})$$

90% interval for the expected revenue increase:

[3471, 5836] dollars.

What does this add to our advice to Andy?

R script · Part 8: Confidence interval for the change

Can advertising keep working at the same rate?

The linear model implies:

$$\frac{\partial E(SALES \mid PRICE, ADVERT)}{\partial ADVERT} = \beta_3$$

Would another 1,000 dollars of advertising have the same effect at every initial advertising level?

What model change could represent diminishing returns?

R script · Part 9: Diminishing returns

A quadratic advertising model

$$SALES_i = \beta_1 + \beta_2 PRICE_i + \beta_3 ADVERT_i + \beta_4 ADVERT_i^2 + e_i \tag{11}$$

```
quadratic_model <- lm(sales ~ price + advert + I(advert^2),
                      data = andy)
```

Nonlinear in advertising; **linear in the coefficients**.

R script · Part 9: Estimate the quadratic model

The advertising curve

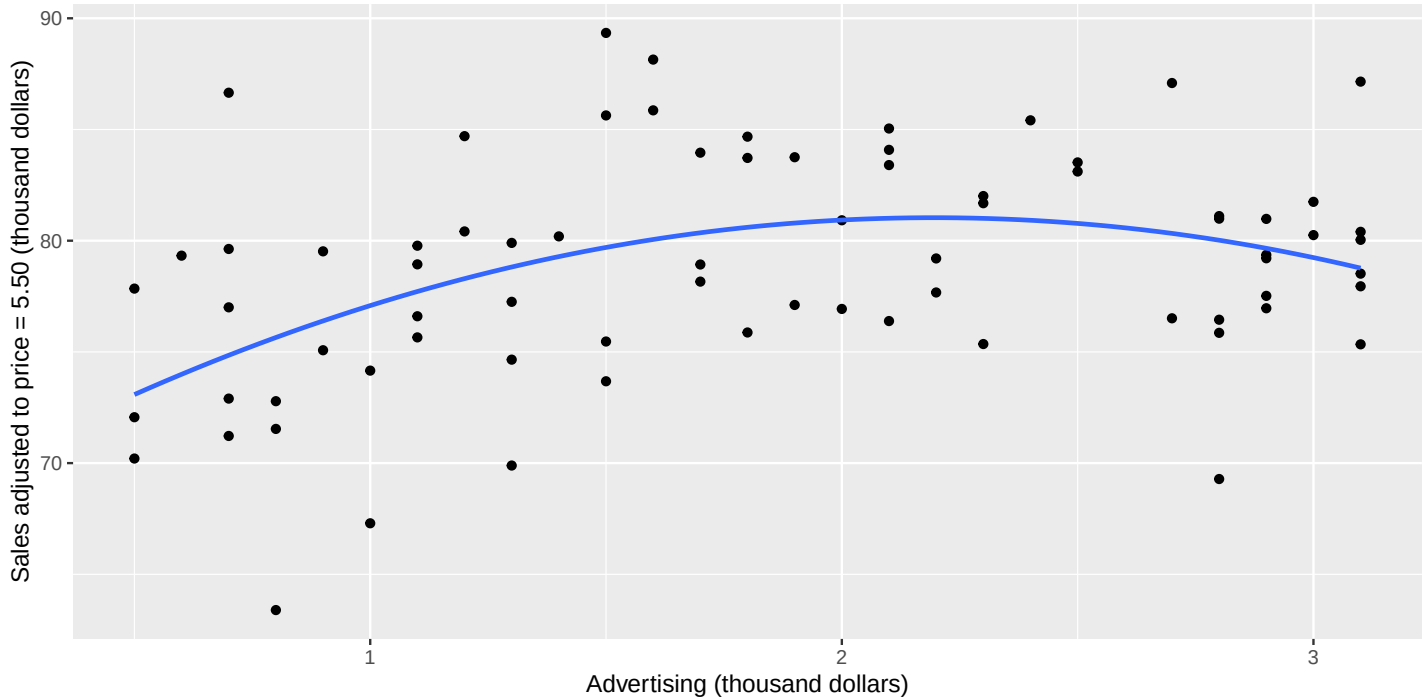


Figure 5: Sales adjusted to a fixed price of 5.50 dollars, with the fitted quadratic advertising curve

Where is an extra advertising dollar most effective?

R script · Part 9: The adjusted advertising plot

A coefficient is no longer the whole marginal effect

Differentiate with respect to advertising, holding price fixed:

$$\frac{\partial E(SALES \mid PRICE, ADVERT)}{\partial ADVERT} = \beta_3 + 2\beta_4 ADVERT \qquad (12)$$

Estimated marginal effect:

$$12.151 - 5.536 ADVERT$$

At $ADVERT = 0.5$: 9.383. At $ADVERT = 2$: 1.079.

R script · Part 9: Calculate marginal effects

When one effect depends on another variable

An **illustrative** hourly wage model:

$$WAGE = 5 + 1\,EDUC + 0.20\,EXPER + 0.05(EDUC \times EXPER) + e$$

$$\frac{\partial E(WAGE)}{\partial EDUC} = 1 + 0.05\,EXPER$$

What is the effect of one more year of education at 0 and 20 years of experience?

R script · Part 10: The interaction example

Andy's decision revisited

- The estimates support a negative price relationship and a positive advertising relationship over the relevant range.
- Advertising significance does not establish cost coverage or profit.
- Diminishing returns make the initial advertising level relevant.

Give Andy a recommendation with one explicit condition and one limitation.

Exit question

A model contains $ADVERT$ and $ADVERT^2$.

An analyst says:

> *"The coefficient on advertising is 12.15, so one additional advertising dollar always brings 12.15 dollars in revenue."*

Correct the interpretation. What must you know before calculating the effect?

Reference material

Same coefficient, but check the standard error

FWL gives the same slope and the same residuals.

But the residual regression sees only **one** estimated coefficient:

Table 5: Conventional standard errors and the degrees-of-freedom correction

Calculation	Residual degrees of freedom	Advertising SE
Full model	72	0.6832
Residual regression, uncorrected	74	0.6739

For inference, use the full model or restore its $N - K$ correction.

R script · Review of Part 5: Compare FWL with the full model

Large samples and further reading

Consistency: with suitable assumptions, estimates become close to the population coefficient with high probability as the sample grows.

Asymptotic normality: large-sample approximations can support inference without normal errors.

A large sample does not automatically solve omitted-variable bias.

Reading: §5.7; Appendix 5A (OLS derivation), 5B (delta method), 5C (simulation), 5D (bootstrap).

Candidate advertising expenditure

Ignoring additional production costs, equate marginal revenue to one dollar per advertising dollar:

$$\beta_3 + 2\beta_4 A^* = 1 \quad \Longrightarrow \quad \widehat{A}^* = \frac{1 - b_3}{2b_4} = 2.014$$

About **2014 dollars per month**.

This is a model-based candidate, conditional on the assumptions and omitted costs.

R script · Presentation support: Candidate advertising level

Sources

Chapter 5, especially §§5.1–5.6 and Examples 5.1–5.15 (Hill et al. 2018).

The companion R script downloads the original Andy observations directly from the publisher and uses tidyverse for data preparation and graphics.

Publisher's Andy data file

The plane illustration reproduces textbook Figure 5.1 from the existing course materials. The remaining plots use the Andy observations.

Hill, R. Carter, William E. Griffiths, and Guay C. Lim. 2018. *Principles of Econometrics*. 5th ed. Wiley.